From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work

Sora Kim

Pharmaceutical Manufacturing Industry, South Korea

sorakim.lab@gmail.com

Abstract Standard operating procedures (SOPs) in pharmaceutical manufacturing environments are legally binding documents, but their practical force depends on whether they are reachable at the moment action is required. This paper argues that SOP accessibility is not only a retrieval problem but an HCI problem about how access conditions shape interpretation upstream of formal task execution. Interface structure determines which procedures become visible and comparable — making retrieval design a tractable target for HCI intervention even within the validated-archive constraint. In GMP-regulated environments, practitioners often work through validated archives that cannot be modified, meaning that retrieval support must operate externally rather than by restructuring source documents. I develop and evaluate an external keyword-searchable retrieval layer over a corpus of 500 synthetic GMP SOPs and show that reducing retrieval burden does more than save time: it expands the procedural landscape practitioners consider available, supportable, and actionable. Findings from a think-aloud pilot study with $N = 20$ GMP-experienced participants show that keyword-based access reduced exploratory retrieval time from approximately six minutes to approximately one minute, that result cues were rated useful by 18 of 20 participants, and that all 20 indicated willingness to reuse the system in live archive conditions. Participants also described a broader shift: when procedures become easier to reach, they become easier to compare, easier to justify using, and less dependent on informal colleague mediation. I further introduce an AI support layer as a conceptual design extension addressing interpretive burden beyond retrieval by explaining why a result appeared, surfacing related procedures, and keeping alternative interpretations visible without resolving them. The paper makes three contributions: it introduces SOP accessibility as a design problem specific to GMP-regulated environments; identifies the validated-archive constraint as a novel design boundary for retrieval research; and proposes an AI support layer as a design direction for addressing interpretive burden beyond retrieval, grounded in pilot findings and established human-AI interaction guidelines, while acknowledging that this layer has not yet been empirically evaluated.

Keywords: SOP accessibility, document retrieval, pharmaceutical manufacturing, GMP, interpretive variability, invisible work, regulated environments, judgment-preserving design

Introduction

Standard operating procedures (SOPs) are among the most consequential documents in regulated manufacturing. In pharmaceutical production environments governed by Good Manufacturing Practice (GMP) regulations, SOPs define how processes must be performed, what counts as acceptable deviation, and what actions can be defended after the fact. Yet the practical challenge of consulting an SOP at the moment it is needed has received little attention in the HCI literature. Prior work on SOP non-compliance has focused on training gaps, motivational barriers, and procedure design rather than on the interaction conditions under which practitioners access documents in the first place (Krockenberger et al., 2008; Hales and Pronovost, 2006). Krockenberger et al. (2008) found that 18.4% of clinical trial coordination staff sought a colleague's guidance rather than retrieving the relevant SOP, concluding that the primary barrier was inadequate retrieval system design. That diagnosis has shaped subsequent system development in regulated environments, yet the design response it implies has not been developed for GMP-specific contexts, where the constraint of validated archives uniquely restricts what retrieval interventions are permissible.

This gap matters because access conditions shape interpretation. When practitioners cannot locate a relevant procedure quickly — because archives are stored as scanned PDFs that block keyword search, because titles are too similar to distinguish, or because time pressure makes trial-and-error retrieval costly — they do not simply fail to comply. They navigate under constrained visibility. They work from memory, defer action, or ask colleagues. These substitutions occur before any formal task error and before any later assignment of responsibility. In this sense, the design of retrieval is not downstream support for already-defined work. It helps determine what procedural possibilities become reachable, comparable, and actionable in the first place.

I argue that document accessibility is an upstream condition on interpretation in regulated work environments. Retrieval burden shapes the range of procedures a practitioner encounters, compares, and can act on with confidence. This argument extends Suchman's (1987) account of situated action to the domain of procedural document use: the formal existence of a procedure does not determine its practical effect; what matters is what is visible and reachable at the moment action is required. In GMP archives, this question becomes a distinct HCI problem because validated source files cannot be modified. Interface structure determines which procedures practitioners can reach and compare — and when that structure fails to externalize the archive's logic, the burden of retrieval falls entirely on the practitioner rather than on the system. Any retrieval improvement must therefore operate externally, over metadata and interaction structure, without altering the documents themselves.

This paper reports a prototype-based pilot study in a pharmaceutical GMP environment. I developed an external retrieval layer — a keyword-searchable metadata index — over a corpus

of approximately 500 procedural documents and evaluated it with $N = 20$ frequent SOP users using a think-aloud protocol. The prototype served simultaneously as a design intervention and as an analytic probe: by observing how participants navigated the system, I was able to make upstream retrieval burden visible in a way that conventional error analysis cannot. Findings show that reducing retrieval effort shifted participant attention from document navigation to document interpretation, and that the friction revealed was structural rather than individual, traceable to archive design rather than user failure.

The paper makes three contributions. First, I introduce SOP accessibility as a design problem within GMP-regulated environments and identify the validated-archive constraint as a novel design boundary for retrieval research — a constraint that prohibits modification of source documents and therefore requires all interventions to operate externally over metadata and interaction structure. Second, I show that retrieval design changes more than navigational efficiency: it changes the interpretive conditions under which practitioners decide what procedures are available to use and justify, with downstream consequences for accountability. Third, I introduce an AI support layer as a conceptual design extension grounded in the limitations identified by the pilot study: not an evaluated system, but a principled design direction that extends the judgment-preservation commitment of the retrieval prototype into the interpretive space that retrieval alone cannot reach.

Related Work

SOP Non-Use and the Limits of Procedural Infrastructure

Prior research has documented that practitioners routinely fail to consult standard operating procedures at the point of action. Krockenberger et al. (2008) found that 18.4% of clinical trial coordination staff sought a colleague's guidance rather than retrieving the relevant SOP, attributing this pattern to inadequate retrieval system design rather than individual non-compliance. A broader literature on procedural adherence confirms that non-compliance with written procedures is rarely a matter of individual negligence; rather, it reflects structural conditions including inaccessible documentation, outdated archives, and the gap between procedures as designed and tasks as performed (Hales and Pronovost, 2006). Yet both lines of work leave a critical question unaddressed: even when retrieval is improved, does better access simply make documents easier to find, or does it reshape the interpretive work that follows? The present study takes up this gap, examining not only whether practitioners can locate SOPs, but how variability in access conditions shapes what they can compare, justify, and act on. The validated-archive constraint in GMP environments adds a further dimension: the design question is not how to restructure the archive, but how to make its logic navigable from the outside.

Professional Document Work and the Persistence of Retrieval Difficulty

The difficulty of locating authoritative documents is not unique to pharmaceutical manufacturing. Sellen and Harper (2002) demonstrate that digital transitions do not resolve the retrieval problems of document-intensive work; rather, they shift the role of documents without eliminating the navigational demands that practitioners face. Harper's (1998) ethnographic study of documentation practices at the IMF shows that documents do not operate as neutral carriers of information: their accessibility structures the work that surrounds them, and gaps in retrieval support create invisible friction that is rarely captured in organizational accounts. Russell-Rose et al. (2018) extend this argument across healthcare, legal, and patent work contexts, demonstrating that retrieval difficulty is a structural feature of professional document environments, not a correctable surface problem. The present study examines how this structural difficulty manifests in GMP contexts, where the constraint of validated archives uniquely restricts what retrieval interventions are permissible — and where retrieval failure has regulatory consequences that extend beyond the navigational experience itself.

Situated Action and the Inaccessible Procedure

The theoretical basis for treating access conditions as consequential for interpretation comes from Suchman's (1987) account of situated action. Suchman argues that human action is not plan-driven but shaped by the immediate conditions of its execution — what is visible, reachable, and actionable in the moment of practice. Applied to SOP use, this framework implies that a procedure's formal existence does not determine its practical effect; what matters is what is accessible to the practitioner at the moment action is required. Unstable or inconsistent access to SOPs thus becomes not a logistical inconvenience but a structural source of interpretive variability, and therefore a design condition on judgment rather than a mere documentation problem. The implication is that retrieval design is accountability design: the procedures practitioners can reach are the procedures they can act on and defend, and the boundary of that set is drawn not by the archive's contents but by the archive's navigability.

Invisible Work and Upstream Labor

When SOPs are inaccessible, practitioners do not simply fail to act; they perform substitutive work — asking colleagues, reconstructing procedures from memory, or deferring action. Star and Strauss (1999) show that work of this kind tends to become invisible: it is not captured by formal indicators, not measured by quality systems, and not attributed to the access failure that produced it. In regulated manufacturing, this invisibility has direct implications: the retrieval burden created by inaccessible documentation sits upstream of any formal error record, and is

therefore structurally absent from CAPA analysis and audit review. The prototype study reported here makes this upstream burden legible — treating it not as individual user difficulty but as a structural condition of the archive. The experience asymmetry in retrieval burden — in which less experienced practitioners bear disproportionate cost because they have not yet internalized the archive's naming logic — is a specific instance of the epistemic access inequality that **?** describes: structural conditions restrict who can effectively navigate the knowledge system, not by competence but by accumulated implicit knowledge the system does not externalize.

Think-Aloud Methods and Prototype-Based Evaluation

The methodological approach in this study draws on an established body of HCI research on think-aloud evaluation and prototype-based inquiry. Ericsson and Simon (1993) provide the theoretical foundation for think-aloud protocols: concurrent verbalization surfaces the cognitive processes underlying task performance without substantially altering their course, making it particularly appropriate for studying navigation and search behavior under realistic task conditions. Nielsen (1994) operationalizes this approach within usability engineering, demonstrating that think-aloud studies with small expert samples reliably surface dominant usability and friction patterns. Jaspers et al. (2004) apply this method in a regulated clinical context directly analogous to the present study: using think-aloud protocols to analyze how clinicians searched paper-based patient records, they developed a prototype interface that reduced retrieval burden without altering source documents — a design constraint that mirrors the validated-file requirement in GMP environments. Hutchinson et al. (2003) establish the logic of the technology probe: a prototype used simultaneously as a design intervention and as an analytic instrument, making it possible to study how access conditions shape behavior rather than simply to evaluate whether a given interface is usable. This dual function is central to the present study: the prototype was not deployed to measure its own effectiveness but to make visible the structural conditions of retrieval burden that conventional error analysis cannot observe.

AI-Assisted Retrieval and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to practitioners navigating complex document environments. Wang and Yin (2021) demonstrate that explanation interfaces alter how decision makers engage with outputs depending on domain expertise — findings that establish retrieval interfaces as active shapers of which procedures become visible and comparable rather than neutral conduits. Green and Chen (2019) show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, even when users are

instructed to exercise independent evaluation. In regulated environments, this finding takes on regulatory force: if AI retrieval support narrows the set of procedures practitioners consider by making AI-surfaced results more salient than unaided search, it reproduces at the retrieval layer the same convergence dynamic that threatens investigation quality at the reasoning layer.

The appropriate response is not to avoid AI support but to design it under a judgment-preservation constraint. Amershi et al. (2019) formalize this constraint as a general principle for human-AI interaction: AI systems should support human control, make their capabilities and limitations clear, and avoid producing outputs that pre-empt human judgment. Applied to retrieval in GMP environments, this principle requires that AI support expand rather than constrain the comparison space: surfacing why a result was returned, making related procedures visible, and flagging interpretive ambiguity without resolving it. The AI support layer introduced in this paper operationalizes this constraint as a design direction, distinguishing judgment-preserving retrieval support from conventional AI approaches that generate recommendations or rank results without explaining their basis.

Prior work has not examined SOP accessibility as a design problem within pharmaceutical GMP environments, where document integrity constraints uniquely restrict retrieval system design. This study addresses that gap directly.

Method

Study Context

This study was conducted in the quality control department of a pharmaceutical manufacturing facility operating under 21 CFR Part 210/211 Good Manufacturing Practice regulations. The procedural archive used in this study consisted of 500 synthetic quality control SOPs generated programmatically in Python. Because actual facility SOPs are subject to non-disclosure obligations, the corpus was constructed to reflect the structural and terminological characteristics of real GMP quality control documentation — including document identifiers, process category conventions, procedural language, and keyword distributions typical of QC operations — without reproducing proprietary content. Documents were rendered as PDFs and parsed to extract metadata for indexing.

This construction approach reflects a domain constraint rather than a methodological preference: GMP procedural documents are proprietary quality records protected under confidentiality obligations that preclude their use in external research without facility authorization. Synthetic generation from structural parameters is the standard approach for research in regulated manufacturing contexts where live data access is unavailable. The validated-file constraint that defines the core design boundary of this work applies equally to synthetic and live corpora:

neither can be modified by the retrieval layer, and both impose the same design requirement that interventions operate externally over metadata rather than by restructuring source documents.

Phase 1: Metadata Mapping and Keyword Analysis

I conducted a systematic analysis of the full procedural archive to construct a searchable metadata index. For each document, I extracted the document identifier, title, process category, and keywords derived from document content. Where keyword coverage was sparse, I supplemented extraction with manual annotation based on process area and task type. I also mapped structural overlap across the corpus: documents sharing similar titles, overlapping keyword sets, or belonging to the same process category were flagged as high-ambiguity clusters. This analysis identified the specific points in the archive where practitioners faced the greatest navigational difficulty — not from individual document complexity, but from the structural similarity of adjacent procedures. The high-ambiguity clusters informed the design of result cues: areas where terminological overlap was greatest required the most contextual differentiation in returned results.

Phase 2: Prototype Development

Based on the metadata index, I developed an external retrieval layer as a Streamlit web application (Figures 1 and 2). The prototype allows practitioners to enter task-relevant keywords — terms describing what they are trying to do, rather than document titles or identifiers — and returns a ranked list of relevant SOPs with document identifiers, process categories, and contextual snippets. Match strength is indicated at three levels (Strong: 5+ matches; Good: 2–4 matches; Weak: 1 match) to support result discrimination before opening documents.

Three design decisions instantiate the judgment-preservation commitment. First, the system decouples retrieval from archive naming conventions: practitioners enter task terms, not document identifiers, which means retrieval does not presuppose the naming knowledge that currently creates asymmetric burden across experience levels. Second, match strength is displayed at three levels rather than as a ranked score: this preserves the practitioner's ability to evaluate result relevance rather than deferring to a numeric ranking whose basis is opaque. Third, the system operates entirely over the metadata index and does not modify, version, or interact with validated source files in any way — maintaining the validated-file boundary that GMP regulation requires.

Phase 3: Pilot Evaluation

I evaluated the prototype with $N = 20$ participants recruited from the same GMP quality control team. All participants were frequent SOP users with daily procedural document search

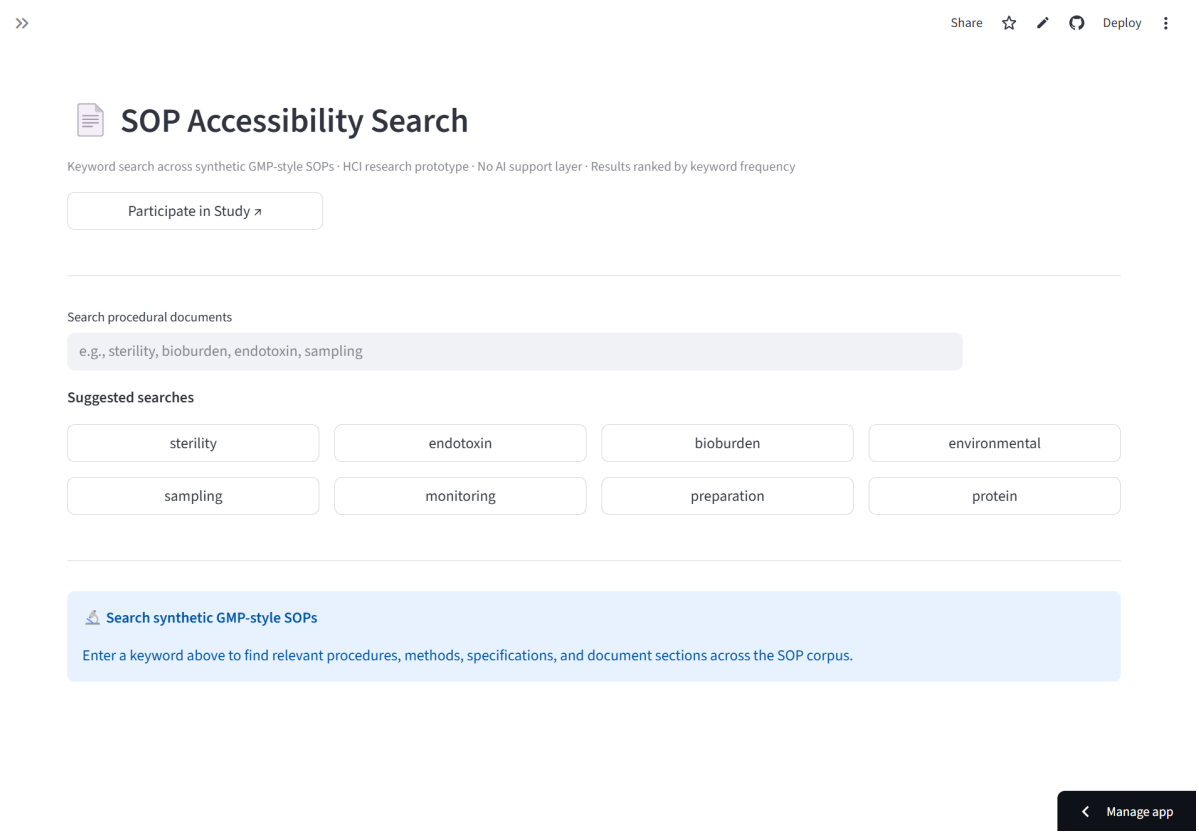


Figure 1: The SOP Accessibility Search prototype at rest, showing the keyword input field and suggested search terms drawn from the synthetic GMP corpus. The interface decouples retrieval from archive naming conventions, allowing practitioners to enter task-relevant terms rather than document identifiers or titles.

experience. Recruiting from a single expert team was a deliberate methodological choice to ensure consistent baseline familiarity with GMP retrieval conventions, prioritizing internal consistency over sample diversity. This design choice may have inflated usability and acceptance ratings relative to a more diverse sample — a limitation acknowledged below.

Participants completed a series of realistic search tasks using the prototype — tasks drawn from their own reported search scenarios — while thinking aloud throughout. Sessions were conducted individually; a researcher observed and took field notes without intervening in participant navigation. Approximate retrieval times were recorded informally as a descriptive measure of navigational effort; these are exploratory observations rather than controlled measurements and are reported as such.

Following the search tasks, participants completed a brief post-task questionnaire assessing ease of finding relevant sections, usefulness of result cues, overall intuitiveness, and willingness to reuse the system in similar work contexts. Open-ended prompts invited participants to describe remaining friction points and to articulate how the prototype differed from their current search practice, with particular attention to changes in what procedures they felt able to consider,

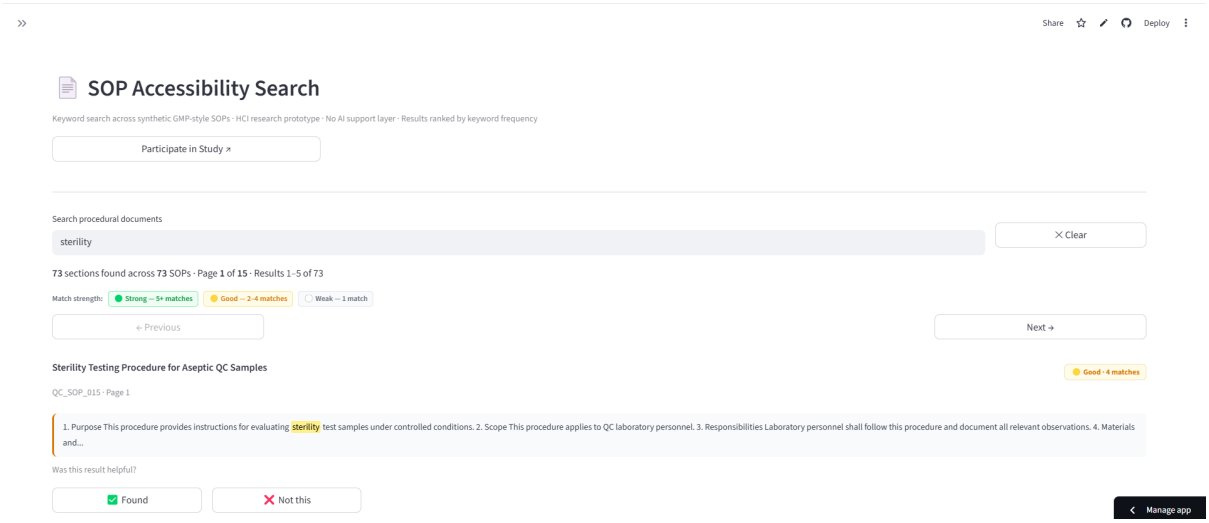


Figure 2: Search results for the query "sterility," returning 73 sections across 73 SOPs. Results are ranked by match strength (Strong / Good / Weak) and include document identifiers, section locations, and contextual snippets with matched terms highlighted.

compare, and justify using.

Think-aloud data were analyzed following Ericsson and Simon's (1993) protocol analysis approach, with particular attention to moments of hesitation, uncertainty, and navigational substitution — instances where participants paused, reworded queries, or expressed doubt about whether a returned result was the correct procedure. These moments are analytically important because they make the interpretive conditions of retrieval visible: they show where the archive's logic is not externalized, and where the burden of disambiguation falls on the practitioner rather than the system.

Findings

Access Burden as an Interpretive Condition

Participants consistently described SOP retrieval as a meaningful burden in their daily work — not a minor inconvenience but a recurring cost that competed with task execution. One participant described the experience directly: "dying of busyness while SOP search is also work." A second framed the burden in terms of its downstream effect on judgment: "when you spend time just finding the right SOP, you are already tired before you even start reading it." A third described the substitution that inaccessibility produces: "when I can't find it quickly, I just use the one I used last time. It's usually close enough." These accounts position retrieval not as a neutral precondition for work but as part of the interpretive conditions under which work begins. The third account is particularly significant: it shows that retrieval difficulty produces not just

slower navigation but a contraction of the procedural landscape practitioners allow themselves to consider.

The prototype reduced this burden in observed use. Exploratory retrieval times dropped from approximately six minutes under current conditions to approximately one minute with the prototype, though these are informal observations rather than controlled measurements. The reduction in time was less consequential, in participants' accounts, than the reduction in what one participant called "navigation anxiety" — the uncertainty about whether the relevant procedure was reachable at all. When the archive's logic is externalized through keyword-based access, practitioners reported that they could begin reading with the confidence that they had reached a relevant procedure, rather than with residual doubt that a more applicable one existed somewhere they could not access. 18 of 20 participants rated the system as intuitive overall; all 20 indicated willingness to reuse the system in live archive conditions.

Accessibility Constraints on Behavioral Repertoire

The most consequential finding concerned the scope of action that practitioners allow themselves under current retrieval conditions. Participants described a systematic pattern: when locating unfamiliar SOPs is difficult, they operate within a personally known subset of the archive, consulting procedures they have used before rather than searching for those that might be more directly applicable. "When you cannot find a SOP easily, you end up acting within the SOPs you already know," one participant explained. Another described the mechanism more precisely: "I know there's probably a better SOP for this, but finding it would take longer than just using the one I'm familiar with. So I use the familiar one and adjust."

In GMP environments, this behavioral adaptation has regulatory consequences. SOPs function as the formal basis for authorized action: they define what practitioners are permitted to do and provide the documented justification for any action taken. When accessibility constraints limit which procedures a practitioner encounters, they also limit the range of authorized actions available — not because the relevant procedures do not exist, but because they are not reachable under realistic task conditions. The resulting interpretive variability is upstream of any formal error and structurally absent from any subsequent error analysis.

The prototype expanded this behavioral repertoire. Participants reported that keyword-based access changed what they felt able to consider, compare, and defend: "I found three procedures I didn't know existed. I've probably been using a workaround for two years when there was a direct procedure the whole time." This account illustrates the mechanism: accessibility interventions do not merely accelerate a pre-existing workflow. They reshape the boundary of what the workflow can contain.

Nomenclature Ambiguity and Onboarding Labor

A second finding concerned the distribution of retrieval burden across experience levels. Senior practitioners described a recurring onboarding burden: new staff, unfamiliar with equipment naming conventions and archive organization, routinely struggle to distinguish between similarly named procedures. A representative example: a senior practitioner directing a new hire to consult the UPLC SOP found that the new hire searched for the HPLC SOP instead, not recognizing the distinction between the two instrument types. Resolving this required senior intervention — "I end up being the SOP search engine for half the team," one senior practitioner noted — creating a pattern of informal knowledge transfer not captured in any formal quality metric.

The prototype partially redistributed this burden. Keyword-based retrieval reduced dependence on exact nomenclature, and contextual snippets helped less experienced users distinguish between adjacent procedures without requiring senior intervention. Participants anticipated measurable reductions in the informal SOP-guidance labor that senior practitioners currently absorb: "if they can find it themselves, I get my afternoon back." This labor is invisible in the sense Star and Strauss (1999) describe: it is not captured by deviation reports, not measured by quality indicators, and not attributed to the archive structure that produces it. Making retrieval navigable without expert mediation moves this category of invisible labor into the design layer, where it can be addressed structurally.

Remaining Friction

Not all friction was resolved. Two categories of residual difficulty emerged consistently. First, no-result states remained ambiguous: when a search returned nothing, participants could not determine whether the issue was a wording mismatch or a genuine gap in archive coverage. Several participants described this ambiguity as more distressing than the original retrieval difficulty: "at least before, I knew I hadn't found it. Now I don't know if it doesn't exist or if I asked the wrong way." A partial design response is to surface empty results as signal rather than failure — indicating that the absence of a result is information about the archive structure, not evidence of user error. The AI support layer introduced below addresses this directly.

Second, 2 of 20 participants reported difficulty discriminating between results when multiple similar procedures appeared in the result set, attributing this to the synthetic corpus's terminological regularity. This residual difficulty is likely reduced in production conditions with messier naming conventions, but it confirms that terminological overlap is a structural feature of GMP archives that retrieval design must address through contextual differentiation, not only through ranked ordering.

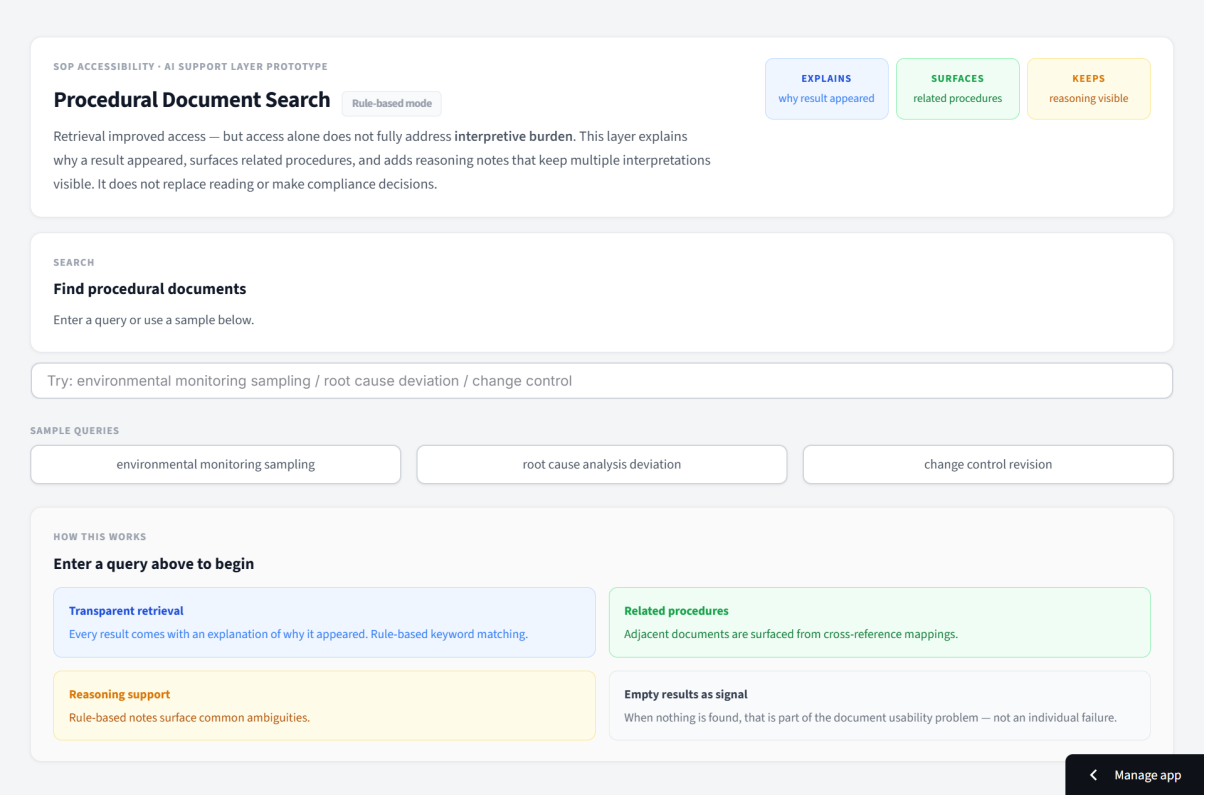


Figure 3: The AI support layer prototype at rest, showing three design principles made explicit in the interface header: EXPLAINS (why a result appeared), SURFACES (related procedures via cross-reference mapping), and KEEPS (reasoning visible by surfacing common ambiguities).

Design Extension: AI Support for Interpretive Burden

The retrieval prototype addressed navigational burden — the cost of locating a relevant procedure. It did not fully address what follows: the interpretive burden of determining which retrieved procedure is most applicable, how its scope relates to adjacent procedures, and where ambiguity in its language may produce inconsistent application. Participants who successfully located relevant SOPs still reported uncertainty about whether they had found the *right* one, and about how to handle cases where two plausible procedures overlapped in scope: "I found it, but I'm not sure this is the one I should be using or the one next to it."

The interpretive burden that retrieval cannot address is structural. Two SOPs may both be returned by a keyword query because both contain the search terms; the question of which applies to the specific task context is not answerable by retrieval logic alone. Resolving it requires either practitioner judgment — which depends on domain knowledge that less experienced practitioners may not yet have — or a support layer that makes the comparison basis available without resolving the comparison. The AI support layer is a design direction for providing the latter.

The AI support layer is organized around three design principles, each of which addresses

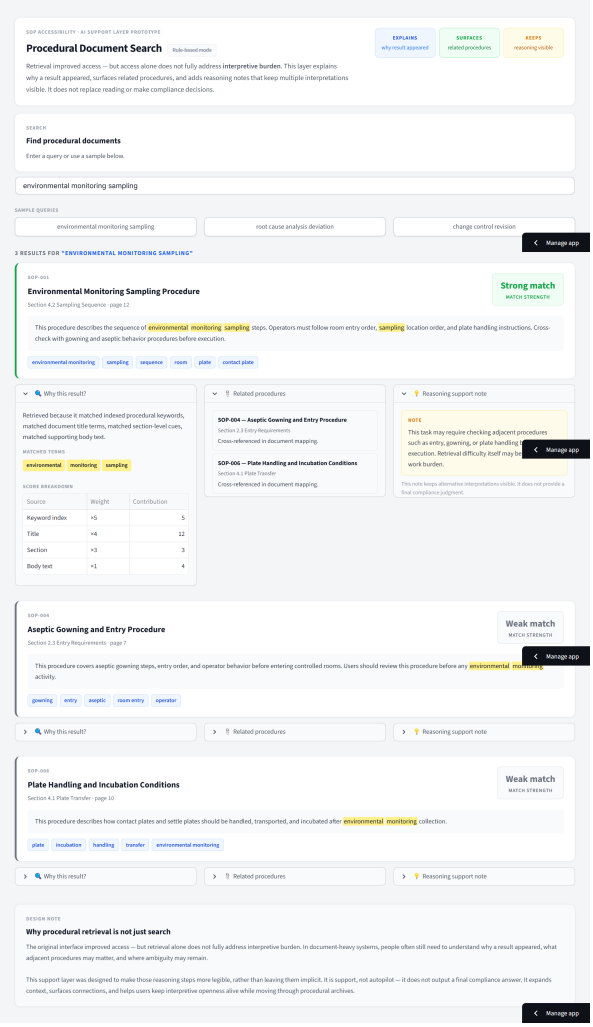


Figure 4: Search results for "environmental monitoring sampling" in the AI support layer, showing a strong-match result (SOP-001) with three expandable panels: *Why this result?* (score breakdown by keyword index, title, section, and body text); *Related procedures* (adjacent documents surfaced from cross-reference mappings); and *Reasoning support note* (rule-based note flagging potential interpretive ambiguity without resolving it).

a distinct component of interpretive burden identified in the pilot evaluation. Each principle instantiates the judgment-preservation constraint (Amershi et al., 2019): the layer expands the cues available to practitioners without producing outputs that function as quality decisions.

Explains why a result appeared. Each returned result includes a transparent account of its retrieval basis — which keywords matched, in which document fields, and at what weight. This addresses the opacity of ranked retrieval outputs. A practitioner who understands why a procedure was returned is better positioned to evaluate whether that basis is relevant to their task, and whether a different query would surface more applicable alternatives. Making retrieval reasoning visible is not merely a usability enhancement; it is a condition for interpretive accountability. When a practitioner can see that a result was returned because it matched on section headers rather than body text, they can make an informed judgment about whether that

match reflects procedural applicability or superficial terminological overlap.

Surfaces related procedures. Each result includes adjacent documents surfaced from cross-reference mappings — procedures structurally connected to the returned result and potentially relevant to the same task. This addresses the scope ambiguity that participants identified: the difficulty of knowing whether a retrieved procedure covers the full range of a task, or whether adjacent procedures define its boundaries. Surfacing related procedures does not resolve this ambiguity; it makes the comparison space available for practitioner evaluation. The design goal is not to identify the correct procedure but to make it possible for the practitioner to identify it themselves, with access to the structural relationships that currently require senior knowledge to navigate.

Keeps interpretive options open. Each result includes a rule-based reasoning support note that surfaces common ambiguities associated with the retrieved procedure — cases where similar tasks are governed by overlapping procedures, or where terminology in the procedure may apply to more than one process context. These notes do not recommend a compliance decision; they flag that multiple interpretations may be viable and that practitioner judgment is required to resolve them. The interface makes this constraint explicit: "this note keeps alternative interpretations visible. It does not replace reading or make compliance decisions."

The layer also addresses the no-result ambiguity identified in the pilot study through an "empty results as signal" design principle: when a query returns no results, the interface indicates that this is information about the archive structure — a possible gap in coverage or a terminology mismatch — rather than evidence of user error. This reframes retrieval failure as a structural finding, moving the interpretation of null results from the practitioner's burden to the system's responsibility.

This layer has not been subjected to empirical evaluation. It is presented here as a principled design direction grounded in the limitations identified by the pilot study and in established human-AI interaction guidelines — not as a validated system. The distinction matters for the paper's contribution: the first two contributions rest on empirical findings; the third rests on design argument, grounded in theory and in the specific constraints of the GMP context. Empirical evaluation of the AI support layer is the primary direction for future work.

Design Principles

The findings and design extensions establish four principles for retrieval system design in compliance-constrained environments. These principles are not specific to pharmaceutical GMP; they apply to any regulated environment in which retrieval systems must operate over validated archives without modifying source documents, and in which the accountability consequences of retrieval failure extend beyond navigational inconvenience.

Decouple retrieval from naming conventions. Retrieval systems that require practitioners to know document identifiers or naming conventions redistribute the cost of archive navigation to practitioners with less accumulated implicit knowledge. Keyword-based access that maps task terms to document content — without requiring practitioners to know what the relevant procedure is called — reduces this asymmetric burden and expands the procedural landscape accessible to less experienced practitioners.

Provide contextual discrimination, not ranked scores. Match strength indicators that provide enough context to distinguish between adjacent procedures — through snippets, category labels, and explanatory cues — preserve practitioner interpretive authority. Numeric scores that rank results without explaining their basis transfer interpretive authority to the retrieval system, which is inappropriate in compliance-constrained environments where the practitioner remains responsible for the procedural judgment.

Surface null results as structural information. When retrieval returns no results, the interface should indicate whether this reflects a wording mismatch or a genuine archive gap. Null results that appear as simple failure states create the impression that the practitioner has searched incorrectly; null results that appear as structural information create the impression that the archive has a gap — which is often the more accurate interpretation and the more actionable one.

Extend comparison-preserving logic into interpretive support. Retrieval tools that help practitioners locate relevant procedures should be extended by support tools that help them compare those procedures and identify interpretive ambiguities. The AI support layer proposed in this paper is one design direction for this extension; its governing principle — expanding the comparison space without resolving the comparison — generalizes to any support layer added above the retrieval layer.

Discussion

Accessibility as a Design Condition on Interpretation

The central finding of this study is not simply that a retrieval tool reduces search time — though it does — but that access conditions shape the interpretive space within which practitioners act. When SOP retrieval is structurally difficult, practitioners do not simply search less efficiently; they act within a personally known subset of the procedural archive, consulting familiar procedures rather than searching for those that might be more applicable. In GMP environments, where SOPs are the formal basis for authorized action, retrieval constraints translate directly into constraints on what a practitioner can justify doing. The interpretive burden therefore begins not at the moment of reading a procedure but upstream, at the moment of deciding which

procedures are reachable — a moment that conventional quality metrics do not capture and that accountability analysis cannot reach.

This finding extends Suchman's (1987) account of situated action. The formal existence of a procedure does not guarantee its practical force; what matters is what is accessible at the moment action is required. The consequences of accessibility constraint are therefore not primarily compliance failures in the conventional sense — violations of documented procedure — but interpretive contractions: practitioners acting within a reduced set of procedural possibilities because the archive does not make the full set reachable. That Krockenberger et al.'s (2008) 18.4% colleague-consultation finding has persisted across decades of procedural system development is consistent with this interpretation: the structural conditions producing retrieval substitution have not been adequately addressed because they have not been treated as an interaction design problem.

Invisible Work Across Experience Levels

The onboarding finding illustrates the mechanism that Star and Strauss (1999) describe as invisible work. Senior practitioners' informal SOP guidance labor is real, recurring, and consequential: it consumes experienced practitioners' time, creates informal dependencies between staff, and shapes how new practitioners develop their procedural knowledge. It is invisible in the specific sense that Star and Strauss identify: it does not appear in any quality metric, is not captured by deviation reports or CAPA records, and is not attributed to the archive structure that produces it.

The prototype partially surfaced this invisibility by making nomenclature disambiguation possible without colleague intervention. The broader implication is that accessibility failure redistributes labor asymmetrically. Practitioners who already know the system's naming conventions can work around poor retrieval infrastructure; those who do not must invest additional search labor or rely on human mediation. Accessibility is therefore not simply a property of documents but part of how epistemic access is distributed across a workforce — and retrieval design is one site at which that distribution can be made more equitable.

Judgment-Preserving Design in Regulated Environments

The AI support layer proposed in Section instantiates the judgment-preservation principle (Amershi et al., 2019) in a specific design form. The layer expands interpretive cues rather than resolves interpretive questions: it makes retrieval reasoning visible, surfaces structural relationships between procedures, and flags ambiguities without recommending how they should be resolved. This design commitment is not merely a normative preference; in GMP environments, it is a regulatory requirement. Any system that produces outputs that function

as compliance decisions — that recommends which procedure to use, determines whether a retrieved result is applicable, or resolves the interpretive ambiguity between two overlapping procedures — introduces a traceability problem for the quality record. The practitioner's judgment must remain attributable and documentable.

The shared logic of the retrieval prototype and the AI support layer is comparison-preserving intervention: both expand the set of alternatives a practitioner can hold in view at different points in the interpretive sequence. Retrieval does this by making a wider range of procedures reachable; the AI layer does this by keeping the comparison space open once procedures have been reached. Neither resolves the interpretive question; both improve the conditions under which the practitioner can resolve it themselves. This design logic is not specific to SOP accessibility — it generalizes to any compliance-constrained context in which practitioners must exercise accountable judgment and the system must support rather than pre-empt that judgment.

Limitations and Future Work

Several limitations bound the interpretation of these findings. The synthetic corpus has less terminological and structural variation than a live archive would, which may have underrepresented navigational difficulty and reduced result discriminability. Participant recruitment from a single team within a single facility introduces potential for inflated usability and acceptance ratings. The study lacks a comparative baseline, as no prior HCI tooling existed in this context. Think-aloud sessions were exploratory and session-based; they do not support claims about long-term adoption, sustained behavior change, or downstream effects on interpretation quality or regulatory compliance.

The AI support layer has not been empirically evaluated. Its three design principles are grounded in pilot findings and in established human-AI interaction guidelines, but their effectiveness in live deployment remains to be established. The most important direction for future work is deployment with a live procedural archive, which would allow both the retrieval prototype and the AI support layer to be evaluated under production conditions — including the full navigational difficulty, time pressure, and interpretive stakes of real GMP document work. Future work should also examine the interpretive consequences of improved access: whether expanded procedural reach leads to more consistent interpretation or introduces new forms of ambiguity by enlarging the comparison set. The AI support layer's comparison-space management is designed to address this in principle; empirical evaluation will determine whether it does so in practice.

Conclusion

This paper introduced SOP accessibility as an upstream condition on interpretive practice in regulated manufacturing environments. When retrieval is structurally difficult, practitioners act within a personally known subset of the procedural archive — not because they lack compliance motivation, but because the archive does not make a broader range of procedures reachable. This constraint shapes what procedures remain visible, comparable, and defensible before any formal task error occurs and before responsibility is later assigned.

The prototype-based pilot study demonstrated that an external retrieval layer — operating over metadata without modifying validated source files — can reduce this upstream burden. Eighteen of twenty participants found the system intuitive, all twenty indicated willingness to reuse it in live archive conditions, and participants described not only faster navigation but an expanded sense of what procedures they could consider, compare, and justify using. The invisible onboarding labor that senior practitioners currently absorb as informal SOP guidance was partially redistributed to the design layer, where it can be addressed structurally rather than managed interpersonally.

Three contributions follow. First, SOP accessibility is introduced as a design problem specific to GMP-regulated environments, where the validated-archive constraint creates a design boundary with no direct precedent in existing retrieval research: all interventions must operate externally over metadata without modifying source documents. Second, the external metadata layer is demonstrated as a viable strategy for reshaping the interpretive conditions of procedure use within compliance constraints — changing not only navigational speed but the boundary of the procedural landscape practitioners allow themselves to consider. Third, the AI support layer is proposed as a judgment-preserving design direction that extends comparison-preserving logic into the interpretive space that retrieval alone cannot reach, making retrieval reasoning transparent, surfacing related procedures, and keeping interpretive options open without substituting for practitioner judgment.

The wider implication is that document accessibility in regulated environments is not a usability problem to be solved once and set aside. It is part of how accountability conditions are produced in practice — determining which procedures practitioners encounter, which alternatives they can compare, and which authorized actions remain within reach.

References

Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Paul Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-

Gil, and Eric Horvitz. Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19. ACM, 2019.

K. Anders Ericsson and Herbert A. Simon. *Protocol Analysis: Verbal Reports as Data*. MIT Press, rev. edition, 1993.

Ben Green and Yiling Chen. The principles and limits of algorithm-in-the-loop decision making. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW):Article 50, 2019.

Brigette M. Hales and Peter J. Pronovost. The checklist — a tool for error management and performance improvement. *Journal of Critical Care*, 21(3):231–235, 2006.

Richard H. R. Harper. *Inside the IMF: An Ethnography of Documents, Technology, and Organizational Action*. Academic Press, 1998.

Hilary Hutchinson, Wendy Mackay, Bo Westerlund, Benjamin B. Bederson, Allison Druin, Catherine Plaisant, Michel Beaudouin-Lafon, Stéphane Conversy, Helen Evans, Heiko Hansen, Nicolas Roussel, and Björn Eiderbäck. Technology probes: Inspiring design for and with families. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '03, pages 17–24. ACM, 2003.

Monique W. M. Jaspers, Thiemo Steen, Cor van den Bos, and Maud Geenen. The think aloud method: A guide to user interface design. *International Journal of Medical Informatics*, 73 (11–12):781–795, 2004.

M. Krockenberger, K. Engel, S. Willmann, and B. Drewelow. Usage and usability of standard operating procedures among coordination centers for clinical trials. *Journal of Clinical Pharmacology*, 48(4):408–414, 2008.

Jakob Nielsen. *Usability Engineering*. Morgan Kaufmann, 1994.

Tony Russell-Rose, Jon Chamberlain, and Farhad Shokraneh. Information retrieval in the workplace: A comparison of professional search practices. *Information Processing & Management*, 54(6):1042–1057, 2018.

Abigail J. Sellen and Richard H. R. Harper. *The Myth of the Paperless Office*. MIT Press, 2002.

Susan Leigh Star and Anselm Strauss. Layers of silence, arenas of voice: The ecology of visible and invisible work. *Computer Supported Cooperative Work*, 8(1–2):9–30, 1999.

Lucy A. Suchman. *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge University Press, 1987.

Xinru Wang and Ming Yin. Are explanations helpful? A comparative study of the effects of explanations in AI-assisted decision-making. In *Proceedings of the 26th International Conference on Intelligent User Interfaces*, IUI '21, pages 318–328. ACM, 2021.